Roll No. ___________

Pandit Deendayal Energy University

Re - End Semester Examination – (May, 2025)
B. Tech. (Computer Engineering)
Semester - VI

Course Name : Big Data Analytics
Course Code : 23CP309T

Date: 09/05/2025
Time: 3 Hours
Max. Marks: 100

Instructions:

1. Do not write anything other than your roll number on question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Question No.	Description	Marks	Course Outcomes (CO's)
Q1-A	**Answer the following (Attempt any one)** 1. For a smart city infrastructure scenario, where do the volume, velocity, and variety of data necessitate the adoption of big data analytic techniques? 2. Draw and explain Hadoop architecture with eco system.	**10**	**CO1**
Q1-B	You are tasked with storing a 1 TB log file in HDFS with the default replication factor of 3. (a) Explain how HDFS stores the file and maintains fault tolerance. Include a diagram of NameNode and DataNode interactions. (b) What happens if one DataNode fails during write? Explain the recovery process. (c) Assume block size is 128 MB. Calculate the number of blocks created and total storage consumed. (d) How is reliability ensured in Hadoop?	**10**	**CO1**
Q.2.	Your team is building a real-time analytics system to process user clicks using Kafka and Spark Streaming. (a) Explain the architecture of Apache Kafka including topics, partitions, brokers, and consumers, Draw a neat architecture diagram. (b) What is the role of the Kafka producer, and how does it achieve high throughput? (c) Describe how Spark Structured Streaming integrates with Kafka and handles fault tolerance using checkpointing and offsets. (d) Assume a Kafka topic has 8 partitions and 4 brokers. If 3 consumer instances in the same consumer group are reading this topic, explain how the partitions are assigned and what happens if one consumer fails.	**20**	**CO6**
Q.3.	**Answer the following:** 1. Compare and contrast Pig Latin with SQL in terms of their syntax, capabilities, and performance for data processing tasks. 2. A Spark job reads data from HDFS into 4 partitions. A repartition(8) operation is then applied, followed by a map() and a reduceByKey(). How many stages will the DAG have, assuming default behavior? 3. Consider a simple social network graph where edges represent "follows" relationships. If User A follows Users B and C, User B follows User C, and User D follows A and C:	**5** **5** **5**	**CO3** **CO3** **CO2**

	i. Draw the graph in pyspark. **ii.** Find the in-degree and out-degree of User C using pyspark? **iii.** Find which user is the most "followed" using pyspark? 4. You are analyzing a large dataset of IoT sensor readings. Which visualization tools or libraries would you choose to create interactive and scalable visualizations? Justify your choices based on the dataset size and the type of insights needed.	**5**	**CO5**
Q.4.	1. For DataFrame df, the df.printSchema() gives the following structure: |-- customer_id: integer (nullable = true) |-- gender: string (nullable = true) |-- age: double (nullable = true) |-- annual_income: double (nullable = true) |-- spending_score: double (nullable = true) |-- membership_years: integer (nullable = true) **Develop Spark SQL queries to perform the following:** **i.** Calculate the **average spending score** of customers aged between 25 and 40 (inclusive). **ii.** Find the **maximum annual income** among customers who have been members for more than 5 years. **iii.** Determine the **percentage of female customers** whose spending score is above 70. **iv.** Compute the **average income** grouped by gender and rounded membership years (e.g., 1–3, 4–6, etc.). 2. List five transformation functions of Spark with suitable example. 3. Discuss different persistence strategies in Spark.	**10** **5** **5**	**CO4** **CO4** **CO4**
Q.5. A.	1. Consider the following directed graph representing web pages and hyperlinks: **Page A links to B and C, Page B links to C, Page C links to A,Page D links to C** Assume a damping factor of 0.85. Perform **one iteration** of the PageRank algorithm starting with equal PageRank values for all nodes. Show intermediate steps and resulting PageRank values after the first iteration.	**10**	**CO2**
Q.5. B.	Explain the stages involved in building a Machine Learning pipeline for processing large-scale textual data. Illustrate your answer with a suitable real-world example (e.g., sentiment analysis or spam detection), and highlight how scalability is achieved using distributed frameworks like Apache Spark.	**10**	**CO5**
	OR		
Q.5. B.	Consider a `patients` collection in a hospital management system. Each document includes: { "patient_id": "P567", "name": "Ravi Sharma", "age": 45, "diagnosis": ["diabetes", "hypertension"], "last_visit": ISODate("2025-03-20"), "status": "active"} **Write MongoDB queries to:** i. Insert a new patient record. ii. Update the status of patients who haven't visited in over a year to `"inactive"`. iii. Find all patients diagnosed with both diabetes and hypertension. iv. Delete records of patients marked as `"inactive"` and whose last visit was before 2023.	**10**	**CO5**